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SHORT-PAPER

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Concept Drift in Bias and Sensationalism Detection: An Experimental Study

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***Abstract*—Due to easy dissemination of news in social media and the Web, there has been an increasing rise of disinformation on important political issues like elections in recent years. Computational solutions for automatic bias and sensationalism detection for news articles can have tremendous impact if used in the right way. Because news is an ever-shifting domain, concept drift is an issue that must be dealt with in any real-world computational news classification system that relies on features and trained machine learning models. Yet, an empirical study of concept drift in such systems, especially popular systems released recently as open-source and used within organizations, has been lacking thus far. This short paper reports results on an empirical study specifically designed to assess concept drift, using an open-source, popular computational news classification system, on real news data crawled from the Web. We find that even a gap of two years (2017 vs. 2019) can lead to significant concept drift, a far narrower gap than observed in traditional machine learning domains, making deployment of pre-trained or openly available computational news classification models an ethically suspect issue.**

***Index Terms*—Bias detection, political news, sensationalism**

I. Introduction

Because of the widespread influence of disinformation and 'fake news' on public perception and media [1], [2], especially pertaining to socio-political events like elections [3], and the ease with which such disinformation can be propagated on the Web, there is an growing need for using technology to combat some of these problems [4], [5]. In an ideal world, a system would simply take an article as input and classify it (possibly on a real-valued scale) as fake or real. However, most articles are far more nuanced, containing cleverly disguised bias and a mix of fact and opinion. Therefore, a more tractable problem in the computational media community is to automatically detect more objective aspects like political bias (e.g., whether an article is right-leaning), satirical cues and sensationalism to profile the news article in a more holistic and well-defined way [6].

Several challenges and initiatives have sought to build systems and datasets to achieve this goal. For example, during the Google 2017 Summer of Code at least one industrial-academic collaboration (described subsequently) built such a system and released it on GitHub, having since gained significant traction and popularity. Similarly, the Fake News Challenge[1] has initially sought to tackle the problem of 'stance detection' rather than focus on systems that outright classify an article as fake. Several stance detection systems are now openly available as a result of the challenge, along with datasets[2].

With the release of such systems, and the evolving nature of news and current affairs, there is a real issue of *concept drift*, since the systems are pre-trained and 'frozen' in place, but used continuously on new datasets. Concept drift is an issue that has been recognized in the machine learning community for several decades [7], [8], and is the problem of a model not performing well because there has been a shift in the distribution of data that renders either the parameters or hyperparameters of a trained system, sub-optimal.

To the best of our knowledge, thus far, there has not been a proper study of concept drift in computational news classification systems[3], even popular systems developed in industry and released in the last two or three years. Unlike ordinary machine learning classifiers, there is reason to suspect that the changing nature of the news, and rising populism, could lead to rapid concept drift in existing models.

To address this need, the primary goal in this paper is to study the issue of concept drift with respect to computational news classification systems. For the purposes of the study, we focus on two popular aspects of such systems, *bias detection* and *sensationalism detection*. Bias detection is the problem of determining whether an article is neutral, right-wing or left-wing, possibly along a numerical scale. Sensationalism detection is the similar problem of assigning an 'objectivity' score to an article.

[1] http://www.fakenewschallenge.org/
[2] https://github.com/FakeNewsChallenge
[3] However, there is far more study on computational news classification itself e.g., see [9].

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We empirically investigate the issue of concept drift by using a popular bias and sensationalism detection classifier designed for political news, built during the Google Summer of Code in 2017 and released on GitHub as open-source[4]. The package was developed by an industrial-academic collaboration[5] (the Computational Linguistics and Psycholinguistics Lab at University of Antwerp with support from textgain[6]). Since publication on GitHub, the project has picked up over 80 stars and is among the most popular open-source projects in the computational news profiling/fake news community. In the next section, we describe the experimental study and results in detail.

II. Experimental Study

A. System

The *bias classifier* has a two-tiered architecture, where first the unbiased articles are filtered out, and then a second model distinguishes between right-wing and left-wing bias. Both models are Logistic Regression classifiers based on lexical bag-of-word n-gram (unigrams, bigrams, trigrams) features, implemented through scikit-learn. The full system takes as input a 2-column CSV file, where the first column corresponds to the headlines and second one corresponds to the article texts. The output is a number between -1 and 1, where -1 is most left-biased, 1 is most right-biased, and 0 is least-biased. For evaluation purposes, scores below 0 are considered "left", above 0 are considered "right", and 0 is considered "least-biased". The *sensationalism classifier* uses a non-linear Support Vector Machine which is implemented through scikit-learn, and is based on the following features (extracted from both headline and article): Part-of-Speech (POS) tags (unigrams and bigrams), frequency counts of each punctuation mark, average sentence length, number of all-cap tokens (excluding common abbreviations, and normalized by length of text), number of words that overlap with a *Pattern Profanity* module that was included with the package, and polarity and subjectivity scores (obtained through a *Pattern Sentiment* module). Like the bias classifier system, it takes as input a 2-column CSV file, where the first column corresponds to the headlines and second one corresponds to the article texts. The output is 1 if the input is categorized as sensationalist, 0 if not.

B. Research Questions

Specific research questions (RQs) that are considered for investigation in this paper are:

RQ1 : What is the extent of concept drift (namely, performance change) of the 'pre-trained'[7] bias classifier and sensationalism classifier on the latest (i.e. 2019) evaluation data compared to earlier (2017)?

RQ2 : Do the classifiers still perform comparably when both trained and evaluated on 2019 data?

RQ3 : Can we trust per-source labels i.e. the assumption that the same label should be assigned to all the articles coming from the same domain (e.g., nytimes.com)? Do the conclusions of RQ1 and RQ2 change if we evaluate with respect to labels assigned to individual articles rather than to entire sources?

C. Data Collection

Similar to the original Github source repository, we used MediaBiasFactCheck.com, a resource that contains more than 2700 media sources and categorizes them into classes such as Right-Center, Left-Center, Least-Biased, Left, etc. We collected and aggregated all of these categorizations into a single CSV of all domains along with the categories assigned by each, any additional comments, etc.

Based on this file, we only picked those domains that clearly exhibit characteristics of a potential category and built two crawlers to crawl and create the 2019 dataset. Note that the original dataset provided in the GitHub source is the 2017 dataset. For some larger news domains, we only crawled certain subcategories. One crawler collected bias news from 133 different domains and the other one collected sensational and objective news from 42 different domains. Before using the data for training, we cleaned the articles, removing articles which have very short body or metadata. Note that the 2017 and 2019 datasets have exactly same class distribution noted in Tables I and II.

TABLE I: Label Distribution for Bias Detection Datasets

(a) tr-X means training dataset in X year

(b) ts-Y means testing dataset in Y year

Label	tr-2017	ts-2017	tr-2019	ts-2019
Left	5,000	1,400	5,000	1,400
Left Center	0	3,274	0	3,274
Least Biased	3,800	1,400	3,800	1,400
Right Center	0	1,863	0	1,863
Right	5,000	1,400	5,000	1,400

TABLE II: Label Distribution for Sensational News Detection Datasets.

(a) tr-X means training dataset in X year

(b) ts-Y means testing dataset in Y year

Label	tr-2017	ts-2017	tr-2019	ts-2019
Objective	8,000	500	8,000	500
Sensationalist	8,000	500	8,000	500

D. Experimental protocol

For both classification systems, we have two sets of training and testing datasets each: 2017 training and testing data, which comes from the original Github repository, and the

[4]https://github.com/clips/news-audit

[5]https://www.newsaudit.net/

[6]https://www.textgain.com/

[7]These classifier models are trained on the original 2017 datasets and released to the public thereafter for use.

2019 training and testing data, which we collected via domain-specific crawling, as briefly described earlier. We maintained the same distributions for the 2017 and 2019 datasets to facilitate uniform analysis and comparison.

To obtain empirical answers to RQ1 and RQ2, we obtained the 2017 and 2019 datasets as described above, and conducted multiple experiments by varying training and test datasets. Specifically, we conducted training on the 2017/2019 training datasets and (respectively) tested on 2017/2019 testing datasets. However, we also conducted experiments where we trained on 2017/2019 training datasets, but tested the models on 2019/2017 testing datasets. The idea behind both methodologies is to measure concept drift of the models in two different respects (covering the scope of RQ1 and RQ2), namely when the year is changed (for both training and testing data), how does it affect performance; similarly, what happens when the years vary between training and testing the model?

TABLE III: Class Distribution of Manual and Heuristic Labels of the 100 Sampled Data Points (from 2017/2019 testing datasets) for Sensational News Detection.

Dataset	Objective	Sensationalist
ts-2017-Manual	49	51
ts-2017-Heuristic	50	50
ts-2019-Manual	55	45
ts-2019-Heuristic	50	50

For bias classifier, we included articles from right-center and left-center sources in the testing, but not training, data since the bias of these sources is usually not evident on a per-article basis. Hence, along with the three classes (right, left and least-biased) that are present in the training data, there are two additional in-between classes that were for evaluation only. In order to be counted as correct for the *recall* metric, right-center can be predicted as either right or least-biased. Similarly, left-center can be predicted as left or least-biased. We do not compute precision individually for these additional classes, since it is not a well-defined problem; instead, to calculate the precision of the least-biased class, least-biased, left-center and right-center classes all count as correct.

RQ3 is based on the data collected for RQ1 and RQ2, and requires manual labeling of each article. In this preliminary study (we leave a fuller study of RQ3 for future work), we only consider RQ3 in the context of the sensationalism classifier. Specifically, we randomly selected 50 objective and 50 sensational articles from (2017/2019) testing datasets, therefore, and obtained 100 data points in total for each testing dataset. We manually labelled these 100 articles as either objective or sensational. We then used 2017 and 2019 pre-trained models to evaluate these 100 data points, and compared the manual labeling results with heuristic labeling results. The class distribution of manual labeling and heuristic labeling is shown in Table III. The manual distributions are slightly skewed, since relabeling the articles results in some 'switches' from objective to sensationalist (and vice versa), but overall, the distributions are stable.

E. Results and Discussion

The results of RQ1 and RQ2 are shown in Tables IV and Table V for the bias detection and sensationlism detection systems respectively. For bias classifier, we regard results of training and testing on 2017 dataset as the standard (akin to a 'benchmark'). Assuming training on 2017 and testing on 2019 data, we note that the precision of left, right and least-biased labels respectively drop by about 6%, 7% and 7%. We suspect that this is because the bias classifier relies on lexical features, which may be specific to the current political climate. In other words, one has to be wary of the training data 'expiring' relatively soon (in less than 2 years) and as a result the accuracy starts decreasing. If we change both the training and testing data to 2019, the precision of left, right and least biased labels all increase, validating the concept-drift hypothesis encapsulated by RQ1; namely, it is not a good idea to be keeping a pre-trained model around except for the very short term[8]. Especially, the least biased label's precision increases by about 12%. Therefore, we can conclude that the bias classifier model is still a good model to be using, but one has to be careful about temporally synchronizing the training and test data.

TABLE V: Results: Sensational News Detection System Trials

(a) tr-X/ts-Y means doing the training on X dataset and testing on Y dataset

		[Objective, Sensationalist]
tr-2017/ts-2017	Precision	[0.9115, 0.9266]
	Recall	[0.928, 0.91]
	F-score	[0.9197, 0.9182]
tr-2017/ts-2019	Precision	[0.5741, 0.6284]
	Recall	[0.7280, 0.46]
	F-score	[0.6419, 0.5311]
tr-2019/ts-2019	Precision	[0.9402, 0.9750]
	Recall	[0.976, 0.938]
	F-score	[0.9578, 0.9561]
tr-2019/ts-2017	Precision	[0.5386, 0.5337]
	Recall	[0.502, 0.5700]
	F-score	[0.5196, 0.5512]

For the sensationalism classifier, compared with the results of training and testing on 2017 dataset, the precision and recall of Objective and Sensationalist classes decrease sharply when training on 2017 and testing on 2019. Similar to the bias classifier, the sensationalism classifier can still work well when we do both training and testing on 2019 dataset because all relevant metrics increase slightly for both sensationalist and objective classes.

[8]An interesting avenue for future work that this opens up is to track the effects of this concept drift in near real-time.

TABLE IV: Results: Bias Detection System Trials

(a) tr-X/ts-Y means doing the training on X dataset and testing on Y dataset

Label	tr-2017/ts-2017			tr-2017/ts-2019			tr-2019/ts-2019			tr-2019/ts-2017		
	P.	R.	F.	P.	R.	F.	P.	R.	F.	P.	R.	F.
Left	0.7015	0.7128	0.7071	0.6419	0.3121	0.4200	0.7595	0.9321	0.8369	0.5896	0.6842	0.6334
Left Center	N/A	0.6114	N/A	N/A	0.1499	N/A	N/A	0.5332	N/A	N/A	0.4880	N/A
Least Biased	0.9613	0.3642	0.5283	0.8901	0.4864	0.6290	0.9892	0.7585	0.8587	0.9486	0.2835	0.4366
Right Center	N/A	0.7214	N/A	N/A	0.9237	N/A	N/A	0.6688	N/A	N/A	0.6317	N/A
Right	0.4555	0.815	0.5844	0.3817	0.8014	0.5171	0.5794	0.9128	0.7088	0.4163	0.5992	0.4913

TABLE VI: Sensationalism Detection by Training on tr-2017 (above) and tr-2019 (below) and Testing on (2017/2019) 100 Sampled Data Points Labeled Heuristically and Manually

System	Precision	Recall	F-Measure
17 Manually	[0.6538, 0.6875]	[0.6938, 0.6470]	[0.6732, 0.6667]
17 Heuristically	[0.9038, 0.9375]	[0.94, 0.90]	[0.9215, 0.9183]
19 Manually	[0.6101, 0.5365]	[0.6545, 0.4888]	[0.6315, 0.5116]
19 Heuristically	[0.661, 0.7317]	[0.78, 0.6]	[0.7155, 0.6593]

TABLE VII: Training on tr-2019

System	Precision	Recall	F-Measure
17 Manually	[0.5454, 0.5535]	[0.4897, 0.6078]	[0.5161, 0.5794]
17 Heuristically	[0.5909, 0.5714]	[0.52, 0.64]	[0.5531, 0.6037]
19 Manually	[0.66, 0.56]	[0.6, 0.6222]	[0.6285, 0.5894]
19 Heuristically	[0.96, 0.96]	[0.96, 0.96]	[0.96, 0.96]

These results together express evidence in support of the finding that concept drift is a very real issue, even with modern datasets, and can start impacting performance even within the confines of a single domain (in this case, the US political landscape and political press) within 1-2 years. Thus, anyone using a pre-trained models should tread with caution. The effects are more pronounced for the sensationalism classifier than for the bias classifier, which may reflect the fact that, compared to 2017, 2019 was more divisive and sensationalist for the press.

The results of RQ3 are shown in Tables VI and VII. When comparing manual vs. heuristic, it is clear that the simplifying assumption made by the original author of this system (namely that sensationalism can be classed for entire sources or domains) is not entirely validated. However, relative conclusions, such as made for RQ1 and 2, still hold, though in a much less dramatic way. For example, consider Table VI. We find that (owing to training on the 2017 dataset), the results on the 2017 vs. 2019 dataset (whether considering manually labeled, or heuristically labeled) show the same evidence for concept drift we saw earlier[9]. Thus, the earlier conclusions are still valid but their extent may need to be re-assessed. In future work, we are working to see if a similar conclusion holds for bias detection as well.

III. Conclusion

In this short paper, we presented an empirical study on the important problems of bias and sensationalism detection for popular press articles, especially covering political news. We found that (possibly owing to tumultuous political trends like rising populism and anti-establishment) concept drift is occurring now at a faster pace than one might expect. Thus, pre-trained models or datasets collected even 1-2 years ago may lead to decreased performance. Since several such pre-trained models are now available online, caution is recommended for journalists and other analysts employing such computational tools. The results also indicate the need for more research on this issue, given that safety, trust and bias have all become important issues in the AI community.

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[9] *17 Manually* has 4% higher F-Measure than *19 Manually*.